

CUSP PERIODS OF MODULAR APÉRY SYSTEMS: ONE RECURRENCE, MANY PERIODS

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ABSTRACT. A modular Apéry system (x, F, Φ) — a Hauptmodul x , a form $F = \sum A_n x^n$ of weight $r - 1$ and an Eisenstein source Φ of weight $r + 1$ — has a companion $B = F \cdot D^{-r} \Phi = \sum B_n x^n$ with $d_n^r B_n$ integral, and B_n/A_n tends to an L -value, the period of B at the singularity of the Picard–Fuchs operator nearest the origin. Every singular point of that operator is the image of a cusp or of a fold, and at every cusp image the companion has, under a regularity condition, a period of its own. We prove that the period at the cusp a/c is the constant term of the Eichler integral $D^{-r} \Phi$ along the radial path to $e(a/c)$, give it in closed form as a finite $\mathbb{Q}(\zeta_M)$ -combination of values $L(r, \chi)$, and prove that the polar coefficient of that expansion is exactly $(2\pi i)^{r+1} a_0(\Phi|\gamma)/(r! c^{r+1})$, so that regularity at a cusp is the vanishing of the constant term of Φ there, together with the vanishing of its constant term at ∞ . The main structural result is an orientation dichotomy: for sources of *inner* orientation $(\psi, 1)$ the real part of the period is $L(\Phi, r)$ at every cusp, because $\operatorname{Re} z/(1-z) = -\frac{1}{2}$ on the whole unit circle, whereas for the *outer* orientation $(1, \varphi)$ the far-cusp period is $L(\Phi, r)$ times a rational factor which is -11 for Apéry’s $\zeta(2)$ row and -3 for the Franel row. On $\Gamma_1(5)$ the inner line carries the $\mathbb{Q}(\sqrt{5})$ -period $\xi = \varphi^5 \operatorname{Im} L(2, \psi_4) - \operatorname{Re} L(2, \psi_4)$ at the near cusp and its Galois conjugate at the far cusp of the conjugate source, and its regularity condition is the identity $\operatorname{Re} L(3, \psi_4) = \varphi^5 \operatorname{Im} L(3, \psi_4)$, which we prove from generalized Bernoulli numbers. A census over the fifteen sporadic rows, verified against the monodromy of the row operators to 100–135 digits, a Fricke-sign criterion for cusp-form sources on Fricke hosts, and the consequence for linear-independence proofs over real quadratic fields — the hypothesis tax of the two-place construction is an orientation theorem — complete the picture.

1. INTRODUCTION

Apéry’s sequences $A_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}$ and their companions B_n satisfy one recurrence, $(n+1)^2 y_{n+1} = (11n^2 + 11n + 3)y_n + n^2 y_{n-1}$, and $B_n/A_n \rightarrow \zeta(2)/5$. Beukers [?] read this modularly: $\sum A_n x^n$ is a weight-one form F on $\Gamma_1(5)$ expanded in the Hauptmodul $x = q \prod (1 - q^n)^{5(\frac{n}{5})}$, and $\sum B_n x^n$ is F times the Eichler integral $D^{-2} \Phi$ of a weight-three Eisenstein series Φ . The same structure underlies every sporadic row of Zagier [?], Almkvist–Zudilin [?] and Cooper [?]: a *source* $\Phi = F \cdot Dx$ of weight $r+1$ (r one more than the weight of F), a companion $B = F \cdot D^{-r} \Phi$ whose coefficients $B_n = \sum_{m \leq n} c(m) m^{-r} e_{n,m}$ have denominators dividing $d_n^r = \operatorname{lcm}(1, \dots, n)^r$, and an Apéry limit which is the critical value $L(\Phi, r)$. The arithmetic and the asymptotics of these companions are the subject of the companion paper [?], whose notation we keep.

The Apéry limit is a statement about *one* singular point of the Picard–Fuchs operator $\mathcal{L}$ of the row: the point x_1 nearest the origin, where $B - \xi F$ is analytic exactly for $\xi = \lim B_n/A_n$. But $\mathcal{L}$ has other singular points, and on the *weight-one hosts* (the second-order rows) every one of them is the image of a cusp of Γ . At each such point the local solutions of $\mathcal{L}$ are u , $u \log(x - x_c)$, and there is again a unique Π_c with $B - \Pi_c F$ analytic whenever the inhomogeneity of B is regular there. This paper is about these *cusp periods*: what they are, when they exist, and how they are related to each other.

The question arose in a concrete Diophantine situation [?, §5]. The fold-regular source space of Apéry's $\zeta(2)$ host is two-dimensional over $K = \mathbb{Q}(\sqrt{5})$: besides Apéry's Φ_D , with limit $\zeta(2)/5$, it contains a source $\Phi_{\text{new}} \in \mathbb{Z}[\varphi][[q]]$ whose limit is the new period

$$\xi = \varphi^5 \operatorname{Im} L(2, \psi_4) - \operatorname{Re} L(2, \psi_4) = 0.65563418884065676633\dots,$$

ψ_4 the odd quartic character mod 5. A linear-independence proof over K in the style of Calegari–Dimitrov–Tang [?] needs a conditional function at *both* real places of K , hence needs to know the periods at the *far* singular point $x_2 = -\varphi^5$ as well as at $x_1 = \varphi^{-5}$. Numerically the far period of Apéry's companion is $-\frac{11}{5}\zeta(2)$, not $\zeta(2)/5$, while the far period of the conjugate source $\sigma\Phi_{\text{new}}$ is exactly $\sigma(\xi)$. The explanation is the orientation of the source, and it is the main theorem of this paper.

Results. Section ?? sets up the objects and defines the period at a cusp. Section ?? contains the analytic core: Theorem ?? expands the Eichler integral along the radial path to a cusp, with an absolutely convergent closed formula for the constant term Π_c^q and the polar coefficient ρ_c ; Theorem ?? identifies ρ_c with the constant term of Φ at the cusp, by Bol's identity; Proposition ?? shows that a non-zero constant term of Φ at ∞ destroys every cusp period. Section ?? proves that Π_c^q is the monodromy period (Theorem ??) and that every cusp period lies in the $\mathbb{Q}(\zeta_M)$ -span of the $L(r, \chi)$ (Corollary ??). Section ?? proves the orientation dichotomy (Theorem ??), derives the Galois equivariance of inner sources (Corollary ??) and treats $\Gamma_1(5)$ in full, including the identity $\operatorname{Re} L(3, \psi_4) = \varphi^5 \operatorname{Im} L(3, \psi_4)$ (Proposition ??). Section ?? is the census of the fifteen rows and its confirmation by monodromy. Section ?? proves the Fricke-sign criterion for cusp-form sources (Theorem ??). Section ?? records what is known at folds, Section ?? the consequence for independence proofs, and Section ?? the open questions. All statements called theorems are proved here; numerical statements carry their digit count and the directory of the scripts that produced them.¹

2. MODULAR APÉRY SYSTEMS AND THE PERIOD AT A CUSP

2.1. Systems, sources, companions. Let $\Gamma \supseteq \Gamma_1(N)$ be a congruence subgroup of genus zero, $x = q + O(q^2)$ a Hauptmodul of Γ with integral q -expansion, and $F = 1 + O(q)$ a holomorphic modular form of weight $k \geq 1$ on Γ (with character if $\Gamma = \Gamma_1(N)$). Put $r := k + 1$ and write $D = q d/dq = \frac{1}{2\pi i} \frac{d}{d\tau}$. The *row* is $A(x) = \sum A_n x^n := F$ as a function of x ; the *canonical source* is $\Phi_{\text{can}} := F \cdot Dx$, a form of weight $k + 2 = r + 1$ with $\Phi_{\text{can}} = q + O(q^2)$. A *source* is any $\Phi \in M_{r+1}(\Gamma)$, holomorphic at all cusps, with expansion $\Phi = c_0 + \sum_{m \geq 1} c(m)q^m$. Its *Eichler integral* and *companion* are

$$(1) \quad \Theta := D^{-r}(\Phi - c_0) = \sum_{m \geq 1} \frac{c(m)}{m^r} q^m, \quad B := F \cdot \Theta = \sum_{n \geq 1} B_n x^n, \quad B_n = \sum_{m=1}^n \frac{c(m)}{m^r} e_{n,m},$$

where $e_{n,m} = [x^n](F(q)q^m) \in \mathbb{Z}$ [?, §2]; thus $d_n^r B_n \in \mathbb{Z}$ whenever $c(m) \in \mathbb{Z}$, and $d_n^r B_n \in \mathcal{O}_K$ whenever $c(m) \in \mathcal{O}_K$. (Θ is the series called Ψ in [?]; we reserve ψ for characters.)

¹Two of the numbers were re-derived independently for this paper by short PARI/GP scripts in `paper/cusp_periods/check/`: the far-cusp period $-\frac{11}{5}\zeta(2)$ of Apéry's $\zeta(2)$ companion and the factor -3 of the Franel row (`01_cuspformula.gp`, 76 digits, with an independent Abel-summation cross-check to 21 digits), and the identity $\operatorname{Re} L(3, \psi_4) = \varphi^5 \operatorname{Im} L(3, \psi_4)$ (`02_L3identity.gp`, 76 digits against `lfun`, and the generalized-Bernoulli formula to 77 digits). The showcase numbers are in `03_showcase.gp`.

2.2. The Picard–Fuchs operator. For a function y on $\mathbb{H}$ set

$$(2) \quad \mathcal{L}y := \frac{1}{\Phi_{\text{can}}} D^r \left(\frac{y}{F} \right).$$

Recall Bol's identity [?]: if g transforms like a form of weight $1-r$ under $\gamma \in SL_2(\mathbb{Z})$, then $D^r g$ transforms like a form of weight $r+1$, $D^r(g|_{1-r}\gamma) = (D^r g)|_{r+1}\gamma$.

Lemma 2.1. (i) $\mathcal{L}$ is a linear differential operator of order r in x with coefficients in $\mathbb{C}(x)$; its solution space is spanned by $F\tau^j$, $0 \leq j < r$, so $\mathcal{L}A = 0$. (ii) For every source Φ ,

$$\mathcal{L}B = \frac{\Phi - c_0}{\Phi_{\text{can}}},$$

and $R_\Phi := \Phi/\Phi_{\text{can}}$ is a rational function of x ; in particular $\mathcal{L}B_{\text{can}} = 1$ for the canonical companion. (iii) R_Φ is regular at the image $x_{\mathfrak{c}}$ of a cusp $\mathfrak{c}$ if and only if $\text{ord}_{\mathfrak{c}} \Phi \geq \text{ord}_{\mathfrak{c}} \Phi_{\text{can}}$, and regular at a fold τ_0 (a critical point of x) if and only if Φ vanishes at τ_0 to the order of Dx .

Proof. (ii) is $D^r \Theta = \Phi - c_0$. Φ/Φ_{can} has weight 0 and is Γ -invariant, hence a rational function of the Hauptmodul; Φ_{can} vanishes exactly where Dx does, which gives (iii). For (i), $D = Dx \cdot \frac{d}{dx}$ shows that $\mathcal{L} = \sum_{j \leq r} a_j(\tau) \frac{d^j}{dx^j}$ with $a_r = (Dx)^{r-1}/F^2$. For $0 \leq i \leq r$ the function x^i/F has weight $-k = 1-r$, so by Bol's identity $D^r(x^i/F)$ has weight $r+1$ and $\mathcal{L}(x^i)$ has weight 0: it is a rational function of x . The triangular system $\mathcal{L}(x^i) = \sum_j a_j i(i-1) \cdots (i-j+1) x^{i-j}$, $i = 0, \dots, r$, then determines $a_0, \dots, a_r$ as rational functions of x . Finally $D^r(\tau^j) = 0$ for $j < r$. $\square$

For the rows of this paper $\mathcal{L}$, cleared of denominators, is the operator of the recurrence, normalized by $\mathcal{L}B_{\text{can}} = 1$; e.g. for Apéry's $\zeta(2)$ row $\mathcal{L} = x(1-11x-x^2)\partial^2 + (1-22x-3x^2)\partial - (3+x)$, and the finite singular points of $\mathcal{L}$ are $x=0$, the images $x_{\mathfrak{c}}$ of the cusps at which x is finite, and the images of the folds. On the weight-one hosts (rows **A–F** of Zagier and $\Gamma_1(5)$) all finite singular points other than 0 are cusp images; on the Fricke hosts of the third-order rows the nearest one is the fold $x(i/\sqrt{N})$ and the far one is a cusp or a second fold.

2.3. The local picture at a cusp, and the period. Let $\mathfrak{c} = \gamma\infty$ with $\gamma = ab$ $cd \in SL_2(\mathbb{Z})$, $c > 0$, be a cusp of Γ of width w , with local parameter $q_{\mathfrak{c}} = e(\tau'/w)$, $\tau = \gamma\tau'$. We assume throughout that x is finite at $\mathfrak{c}$ and that the cusp is *simple* for x : $x - x_{\mathfrak{c}} = a_1 q_{\mathfrak{c}} + O(q_{\mathfrak{c}}^2)$, $a_1 \neq 0$. Write

$$u := F|_k \gamma = \phi_0 + O(q_{\mathfrak{c}}), \quad \Phi|_{r+1} \gamma = a_0(\Phi|\gamma) + \sum_{m \geq 1} a_m(\gamma) q_{\mathfrak{c}}^m,$$

and assume $\phi_0 \neq 0$ (F does not vanish at the cusp). Then u is an analytic function of x near $x_{\mathfrak{c}}$, and $\tau' = \frac{w}{2\pi i} \log q_{\mathfrak{c}} = \frac{w}{2\pi i} (\log(x - x_{\mathfrak{c}}) + \text{analytic})$.

Lemma 2.2 (Bol expansion of the companion). *Suppose $c_0 = a_0(\Phi, \infty) = 0$. There is a polynomial P_γ of degree $\leq r-1$ such that, with $\Theta_\gamma := \sum_{m \geq 1} a_m(\gamma)(w/m)^r q_{\mathfrak{c}}^m$,*

$$(3) \quad F(\gamma\tau') = (c\tau' + d)^{r-1} u(\tau'), \quad B(\gamma\tau') = u(\tau') \left[a_0(\Phi|\gamma) \frac{(2\pi i \tau')^r}{r!} + P_\gamma(\tau') + \Theta_\gamma(\tau') \right].$$

If $c_0 \neq 0$, the second formula acquires the additional term $-c_0 u(\tau')(c\tau' + d)^{r-1} (2\pi i \gamma\tau')^r / r!$.

Proof. $D^r \Theta = \Phi - c_0$. If $c_0 = 0$, Bol's identity gives $D^r(\Theta|_{1-r}\gamma) = \Phi|_{r+1}\gamma$, and D^{-r} of $a_0(\Phi|\gamma) + \sum a_m q_{\mathfrak{c}}^m$ is $a_0(\Phi|\gamma)(2\pi i \tau')^r / r! + \Theta_\gamma$ (since $Dq_{\mathfrak{c}}^m = (m/w)q_{\mathfrak{c}}^m$) up to a polynomial of degree $< r$; so $(c\tau' + d)^{r-1} \Theta(\gamma\tau') = a_0(\Phi|\gamma)(2\pi i \tau')^r / r! + P_\gamma + \Theta_\gamma$. Multiply by $F(\gamma\tau') = (c\tau' + d)^k u = (c\tau' + d)^{r-1} u$. If $c_0 \neq 0$, apply this to $\Theta + c_0(2\pi i \tau)^r / r!$, whose r -th derivative is Φ . $\square$

Since $u(x_c) = \phi_0 \neq 0$ and $\log(x - x_c)$ is transcendental over the analytic functions at x_c , (??) says that near a cusp image every solution of $\mathcal{L}y = R$ with R regular is $y_{\text{part}} + u \cdot Q(\tau')$, $\deg Q \leq r - 1$, y_{part} analytic, and $A = F = u \cdot (c\tau' + d)^{r-1}$.

Definition 2.3. Φ (or B) has a period at x_c if there is $\Pi \in \mathbb{C}$ with $B - \Pi F$ analytic at x_c ; Π is then unique and is denoted $\Pi_c(\Phi)$. The *log-coefficient ratio* $\ell_c(B)/\ell_c(A)$ is the quotient of the coefficients of $u\tau'^{r-1}$ in B and in A , i.e. of the top coefficients of the polynomials in (??); equivalently, it is the ratio of the monodromy differences $(M_c - 1)B$ and $(M_c - 1)A$ when $r = 2$.

Proposition 2.4. Let $c_0 = 0$. (i) If $a_0(\Phi|\gamma) \neq 0$ there is no period at x_c . (ii) If $a_0(\Phi|\gamma) = 0$ and $r = 2$ there is a period, $\Pi_c = \ell_c(B)/\ell_c(A) = p_1/c$ where $P_\gamma(\tau') = p_1\tau' + p_0$. (iii) If $a_0(\Phi|\gamma) = 0$ and $r \geq 3$ there is a period iff $P_\gamma(\tau') - \Pi(c\tau' + d)^{r-1}$ is constant for some Π , and then $\Pi_c = \Pi = \ell_c(B)/\ell_c(A)$. At the singularity x_1 nearest the origin the period is the Apéry limit $\lim B_n/A_n$, and $A_n\Pi - B_n$ decays geometrically.

Proof. $B - \Pi F = u[a_0(\Phi|\gamma)(2\pi i\tau')^r/r! + P_\gamma(\tau') - \Pi(c\tau' + d)^{r-1}] + u\Theta_\gamma$, and $u\Theta_\gamma$ is analytic; the bracket is a polynomial in τ' , and $u \cdot (\text{polynomial in } \tau')$ is analytic at x_c iff the polynomial is constant. Only for $r = 2$ is every linear polynomial of the form $\Pi(c\tau' + d) + \text{const}$. The last sentence is the transfer theorem for a function analytic in $|x| < |x_1| + \varepsilon$ except for a single logarithmic singularity at x_1 [?]. $\square$

For $r \geq 3$ the condition in (iii) is the statement that the *period polynomial* of Φ at $\mathbf{c}$ is proportional to $(c\tau' + d)^{r-1}$; on the Fricke hosts of the third-order rows it is automatic at the near fold once $a_0 = 0$ (Theorem ?? below and [?, Thm. B]).

Remark 2.5. Lemma ??(iii) says when R_Φ is regular at x_c . On a weight-one host $\Phi_{\text{can}} = F D x$ vanishes to order exactly 1 at a simple cusp where $F \neq 0$, so R_Φ is regular there iff $a_0(\Phi|\gamma) = 0$: the regularity of the inhomogeneity and condition (i) of Proposition ?? coincide. If R_Φ has a simple pole at x_c , B contains a $\log^2(x - x_c)$ term and no period exists, as in (??) with $a_0 \neq 0$.

3. THE EXPANSION AT A CUSP

3.1. Orientation. The sources of the sporadic rows are finite combinations $\Phi = \sum_d c_d E_{r+1}^{\psi, \varphi}(d\tau)$ of the Eisenstein series

$$E_{r+1}^{\psi, \varphi} = c_0 + \sum_{m \geq 1} \left(\sum_{e|m} \psi(m/e) \varphi(e) e^r \right) q^m,$$

ψ, φ Dirichlet characters (or, on $\Gamma_1(5)$, real combinations of characters, i.e. periodic functions) with $\psi\varphi(-1) = (-1)^{r+1}$ [?]; φ is the character called χ in [?], and $L(\Phi, s) = \sum_d c_d d^{-s} L(s, \psi) L(s - r, \varphi)$. Writing $m = fe$,

$$(4) \quad \Theta = \sum_d c_d d^{-r} \sum_{f \geq 1} \frac{\psi(f)}{f^r} \Lambda_\varphi(q^{df}), \quad \Lambda_\varphi(z) := \sum_{e \geq 1} \varphi(e) z^e = \frac{N_\varphi(z)}{1 - z^Q}, \quad N_\varphi(z) = \sum_{a=1}^Q \varphi(a) z^a,$$

Q the period of φ . We call the source *inner* if $\varphi = \mathbf{1}$, when $\Lambda(z) = z/(1 - z)$ and Θ is a twisted Lambert series, and *outer* if $\psi = \mathbf{1}$, when $\Theta = \sum_d c_d d^{-r} \sum_e \varphi(e) \text{Li}_r(q^{de})$ is a combination of polylogarithms. The poles of Λ_φ are the Q -th roots of unity z_0 with $N_\varphi(z_0) \neq 0$, all simple; for $\varphi \neq \mathbf{1}$ the point $z_0 = 1$ is removable, $\Lambda_\varphi(1) = -\frac{1}{Q} \sum_a a \varphi(a) = -B_{1, \varphi}$. Put

$$w_\varphi(z_0) := \begin{cases} \frac{1}{2} N_\varphi(z_0) - \frac{1}{Q} z_0 N'_\varphi(z_0) & z_0^Q = 1, \\ \Lambda_\varphi(z_0) & \text{otherwise,} \end{cases} \quad n_\varphi(z_0) := \begin{cases} N_\varphi(z_0)/Q & z_0^Q = 1, \\ 0 & \text{otherwise;} \end{cases}$$

for $\varphi = \mathbf{1}$ this is $w(1) = -\frac{1}{2} = \zeta(0)$, $n(1) = 1$.

Theorem 3.1 (expansion at a cusp). *Let $r \geq 2$, $\mathfrak{c} = a/c$ with $\gcd(a, c) = 1$, $\zeta = e(a/c)$, and $q = \zeta e^{-t}$, $t \rightarrow 0^+$. Then*

$$(5) \quad \Theta(\zeta e^{-t}) = \frac{\rho_{\mathfrak{c}}}{t} + \Pi_{\mathfrak{c}}^q + O(t \log \frac{1}{t}), \quad \rho_{\mathfrak{c}} = \sum_d c_d d^{-r-1} \sum_{f \geq 1} \frac{\psi(f)}{f^{r+1}} n_{\varphi}(\zeta^{df}), \quad \Pi_{\mathfrak{c}}^q = \sum_d c_d d^{-r} \sum_{f \geq 1} \frac{\psi(f)}{f^r} w_{\varphi}(\zeta^{df}),$$

both series converging absolutely. Equivalently, for the twisted Dirichlet series $A_{\mathfrak{c}}(s) = \sum_m c(m) e(am/c) m^{-s}$ one has $\Pi_{\mathfrak{c}}^q = A_{\mathfrak{c}}(r)$ and $\rho_{\mathfrak{c}} = \text{Res}_{s=r+1} A_{\mathfrak{c}}(s)$.

Proof. Fix d and f and put $z_0 = \zeta^{df}$, $u = dft$, so $q^{df} = z_0 e^{-u}$. If $z_0^Q \neq 1$, $\Lambda_{\varphi}(z_0 e^{-u}) = \Lambda_{\varphi}(z_0) + O(u)$, and $|\Lambda_{\varphi}(z_0 e^{-u})|$ is bounded uniformly in $u > 0$ and in z_0 , because z_0 runs over the finitely many c -th roots of unity and, for $0 < s \leq 1$ and a root of unity $\zeta' \neq 1$, $|1 - s\zeta'| \geq \min(1, |\text{Im } \zeta'|) > 0$. If $z_0^Q = 1$, then $N_{\varphi}(z_0 e^{-u}) = N_{\varphi}(z_0) - uz_0 N'_{\varphi}(z_0) + O(u^2)$ and $1 - e^{-Qu} = Qu(1 - \frac{1}{2}Qu + O(u^2))$, whence

$$\Lambda_{\varphi}(z_0 e^{-u}) = \frac{N_{\varphi}(z_0)}{Qu} + \left[\frac{N_{\varphi}(z_0)}{2} - \frac{z_0 N'_{\varphi}(z_0)}{Q} \right] + O(u) = \frac{n_{\varphi}(z_0)}{u} + w_{\varphi}(z_0) + O(u),$$

and $\Lambda_{\varphi}(z_0 e^{-u}) - n_{\varphi}(z_0)/u$ is bounded uniformly in $u > 0$ (it is a combination of $1/(e^{Qu} - 1) - 1/(Qu)$ and bounded terms). Insert in (??): the sum over f of $\psi(f) f^{-r} [\Lambda_{\varphi}(\zeta^{df} e^{-dft}) - n_{\varphi}(\zeta^{df})/(dft)]$ converges absolutely and uniformly, so by dominated convergence it tends to $\sum_f \psi(f) f^{-r} w_{\varphi}(\zeta^{df})$, with error $\sum_f f^{-r} \min(ft, 1) \ll t \log \frac{1}{t}$ for $r = 2$ and $\ll t$ for $r \geq 3$; the subtracted terms give $\rho_{\mathfrak{c}}/t$. The Mellin form: $\Theta(\zeta e^{-t}) = \frac{1}{2\pi i} \int \Gamma(s) A_{\mathfrak{c}}(s+r) t^{-s} ds$, and shifting the contour past the pole of $A_{\mathfrak{c}}(s+r)$ at $s = 1$ (of residue $\rho_{\mathfrak{c}}$, from the pole of $L(s-r, \cdot)$ at $s = r+1$) and the pole of Γ at $s = 0$ identifies $\Pi_{\mathfrak{c}}^q = A_{\mathfrak{c}}(r)$. $\square$

Corollary 3.2 (finite form). *$w_{\varphi}(\zeta^{df})$ depends on f only modulo $c_d := c/\gcd(c, d)$ and $\psi(f)$ only modulo the period Q_{ψ} ; with $M = \text{lcm}(c_d, Q_{\psi})$,*

$$(6) \quad \sum_{f \geq 1} \frac{\psi(f) w_{\varphi}(\zeta^{df})}{f^s} = M^{-s} \sum_{j=1}^M \psi(j) w_{\varphi}(e(adj/c)) \zeta_H(s, j/M),$$

so $\Pi_{\mathfrak{c}}^q$ and $\rho_{\mathfrak{c}}$ are finite combinations of Hurwitz values. In the outer orientation ($\psi = \mathbf{1}$, Q the period of φ) this reads

$$\Pi_{a/c}^q = \sum_{j=1}^P \varphi(j) \text{Li}_r(e(aj/c)) \left(\frac{1}{2} - \frac{j}{P} \right), \quad \rho_{a/c} = \frac{1}{P} \sum_{j=1}^P \varphi(j) \text{Li}_{r+1}(e(aj/c)), \quad P = \text{lcm}(c, Q),$$

for a single block $d = 1$; the polar condition involves Li_{r+1} , and Π^q reduces to the Abel-type expression $-\frac{1}{P} \sum_j j \varphi(j) \text{Li}_r(e(aj/c))$ only when $\sum_j \varphi(j) \text{Li}_r(e(aj/c)) = 0$.

Proof. Periodicity is clear; (??) is the definition of ζ_H . For the outer form write $A_{a/c}(s) = \sum_{e,f} \varphi(e) e^r e(aef/c) (ef)^{-s} = P^{r-s} \sum_{j=1}^P \varphi(j) \text{Li}_s(e(aj/c)) \zeta_H(s-r, j/P)$ (group e modulo P) and use $\zeta_H(0, x) = \frac{1}{2} - x$, $\text{Res}_{s=1} \zeta_H(s, x) = 1$. $\square$

Computer-assisted verification 3.3 (`lattice/cusp_periods/01_check gp`, `check/01_cuspformula gp`).

For Apéry's Φ_D ($\varphi = (1, -2, 2, -1, 0) \bmod 5$, $r = 2$), (??) gives $\Pi_0^q = \zeta(2)/5$ and $\Pi_{1/2}^q = -\frac{11}{5} \zeta(2)$ to 76 digits; direct Abel summation of $\sum_{m \leq 4 \cdot 10^4} c(m) m^{-2} (-1)^m e^{-mt}$ at $t = 0.0025, 0.00125$ with one Richardson step reproduces the second to 21 digits. For Φ_{new} the formula gives ξ at the cusps 0 and $1/2$ to 76 digits.

3.2. The polar part is the constant term.

Theorem 3.4. *Let $c_0 = 0$ and $\gamma = ab$
 $cd \in SL_2(\mathbb{Z})$ with $\gamma\infty = a/c$, $c > 0$. Then*

$$(7) \quad \rho_{a/c} = \frac{(2\pi i)^{r+1}}{r! c^{r+1}} a_0(\Phi|_{r+1}\gamma), \quad \text{and} \quad \Pi_{a/c}^q = \frac{p_{r-1}}{c^{r-1}} - \frac{(2\pi i)^r d}{(r-1)! c^r} a_0(\Phi|_{r+1}\gamma),$$

where p_{r-1} is the leading coefficient of the polynomial P_γ of Lemma ???. The error term in (??) is in fact $O(t)$.

Proof. Along $\tau = a/c + iy$ one has $\tau = \gamma\tau'$ with $X := c\tau' + d = i/(cy)$ and $\tau' = (X - d)/c$; here $t = 2\pi y$, so $X = 2\pi i/(ct)$, and $q_c = e(\tau'/w)$ is exponentially small in $1/t$. By Lemma ??,

$$\Theta(\gamma\tau') = X^{1-r} \left[a_0(\Phi|\gamma) \frac{(2\pi i)^r (X-d)^r}{c^r r!} + P_\gamma\left(\frac{X-d}{c}\right) + \Theta_\gamma \right] = a_0(\Phi|\gamma) \frac{(2\pi i)^r}{c^r r!} (X-d)^r + \frac{p_{r-1}}{c^{r-1}} + O(X^{-1}),$$

since $(X-d)^r X^{1-r} = X - rd + O(X^{-1})$, $P_\gamma((X-d)/c)X^{1-r} = p_{r-1}c^{1-r} + O(X^{-1})$, and $\Theta_\gamma X^{1-r}$ is exponentially small. Substituting $X = 2\pi i/(ct)$ and comparing with (??) gives both formulas and the error term $O(X^{-1}) = O(t)$. $\square$

Corollary 3.5. (i) *In the inner orientation, $\rho_{a/c} = \sum_d c_d d^{-r-1} \psi(c_d) c_d^{-r-1} L(r+1, \psi)$, $c_d = c/\gcd(c, d)$; it vanishes automatically if $\gcd(c, \text{cond } \psi) > 1$ and otherwise is a critical-value condition. (ii) *In the outer orientation $\rho_{a/c} \neq 0$ requires ζ^{df} to reach a pole of Λ_φ , i.e. $N_\varphi(\zeta^{df}) \neq 0$ for some d, f .**

Proof. (i) $n(\zeta^{df}) = 1$ iff $c \mid df$ iff $c_d \mid f$; write $f = c_d g$. (ii) is the definition of n_φ . $\square$

Computer-assisted verification 3.6 (`lattice/cusp_periods/04_polar.gp`, 77 digits). Identity (??) was checked against PARI's `mfslashexpansion` at every cusp of $\Gamma_0(N_\Phi)$ for the sources of rows **A**, **C**, **E**, γ, ζ (e.g. **A** at $1/3$: $a_0 = -\frac{1}{8}$, $\rho = 0.57419030889444\dots i$; **E** at $1/2$: $a_0 = -i/16$; γ at $1/2, 1/3$: $a_0 = -\frac{1}{9}, \frac{1}{4}$), all differences $\leq 5.6 \cdot 10^{-77}$. On $\Gamma_1(5)$ the constant-term table of $\Phi_D, \Phi_{\text{new}}$ (`lattice/gamma15/task2/`, ≥ 78 digits) reproduces the polar coefficients of (??) cusp for cusp.

3.3. The third regularity condition.

Proposition 3.7. *If $c_0 = a_0(\Phi, \infty) \neq 0$, then B has no period at any cusp image x_c with $\mathfrak{c} \neq \infty$: for every Π , $B - \Pi F$ contains the non-analytic term $(-1)^{r+1} c_0 (2\pi i)^r c^{-r} u(\tau')/(r! (c\tau' + d))$, which behaves like $u/\log(x - x_c)$, and B_n/A_n converges at best logarithmically. Consequently the space of sources with a period at a given cusp image on a weight-one host is $\{a_0(\Phi, \infty) = 0\} \cap \{a_0(\Phi|\gamma) = 0\}$.*

Proof. By Lemma ?? the extra term is $-c_0 u (c\tau' + d)^{r-1} (2\pi i \gamma \tau')^r / r!$, and $\gamma\tau' = a/c - 1/(c(c\tau' + d))$, so $(c\tau' + d)^{r-1} (\gamma\tau')^r = \sum_{j=0}^r \binom{r}{j} (a/c)^{r-j} (-1/c)^j (c\tau' + d)^{r-1-j}$: a polynomial of degree $r-1$ in τ' plus the term $j=r$, $(-1)^r c^{-r} (c\tau' + d)^{-1}$. The polynomial part can be absorbed into P_γ ; the $(c\tau' + d)^{-1}$ term cannot be cancelled by $\Pi F = \Pi u(c\tau' + d)^{r-1}$ nor by an analytic function. Coefficientwise, $[x^n] u/\log(1 - x/x_c) \asymp x_c^{-n}/(n \log^2 n)$ against $A_n \asymp x_c^{-n}/n$ at the nearest singularity. $\square$

The reason, in the language of Lemma ??: $\mathcal{L}B = (\Phi - c_0)/\Phi_{\text{can}}$, and while Φ/Φ_{can} is a modular function, c_0/Φ_{can} has weight $-(r+1)$ and is not; it is singular at every cusp image where Φ_{can} vanishes. On $\Gamma_1(5)$ this settles a discrepancy noticed in the census: the two outer directions $R_1 = 2E_3^{\mathbf{1}, \text{Re } \psi_4}$ and $R_2 = -2E_3^{\mathbf{1}, \text{Im } \psi_4}$ have $\rho = 0$ at the cusps 0 and $1/2$ and q -side values $\frac{6}{5}\zeta(2)$, $-\frac{2}{5}\zeta(2)$ at 0, yet B_n/A_n misses these values by defects in the exact ratio $-2 = c_0(R_1)/c_0(R_2)$ (`lattice/cusp_periods/00_disc.gp`, exact rationals to $n = 300$, ratio

to 80 digits), the inhomogeneity of the recurrence grows like φ^{5n} , and only the combination $\frac{1}{2}R_1 + R_2 = \Phi_D$ with $c_0 = 0$ has a period. The fold-regular subspace of $M_3^{\text{Eis}}(\Gamma_1(5))$ is therefore two-dimensional (4 Eisenstein directions, 2 independent constant-term functionals).

4. THE PERIOD FORMULA AND THE FIELD OF THE PERIODS

Theorem 4.1 (cusp periods). *Let $a_0(\Phi, \infty) = 0$ and let $\mathfrak{c} = a/c$ be a simple cusp with $x_{\mathfrak{c}}$ finite and $\rho_{\mathfrak{c}} = 0$, i.e. $a_0(\Phi|\gamma) = 0$. Then the log-coefficient ratio of B at $x_{\mathfrak{c}}$ is the constant term of Theorem ??,*

$$\frac{\ell_{\mathfrak{c}}(B)}{\ell_{\mathfrak{c}}(A)} = \Pi_{\mathfrak{c}}^q = A_{\mathfrak{c}}(r) = \sum_d c_d d^{-r} \sum_{f \geq 1} \frac{\psi(f)}{f^r} w_{\varphi}(\zeta^{df}),$$

and whenever B has a period at $x_{\mathfrak{c}}$ (always for $r = 2$), $\Pi_{\mathfrak{c}}(\Phi) = \Pi_{\mathfrak{c}}^q$. If $\rho_{\mathfrak{c}} \neq 0$ the number $A_{\mathfrak{c}}(r)$ still exists but is not a period.

Proof. By (??) with $a_0(\Phi|\gamma) = 0$, $\Pi_{\mathfrak{c}}^q = p_{r-1}/c^{r-1}$, and p_{r-1}/c^{r-1} is the log-coefficient ratio by (??): the coefficient of $u\tau'^{r-1}$ is p_{r-1} in B and c^{r-1} in A . Proposition ?? identifies it with the period. $\square$

Corollary 4.2 (field of the periods). *Every cusp period of an Eisenstein source with characters ψ, φ is a $\mathbb{Q}(\zeta_M)$ -linear combination of the values $L(r, \chi)$, χ ranging over the Dirichlet characters of conductor dividing $M = \text{lcm}(c, \text{cond } \varphi, \text{cond } \psi)$. For $\varphi = \mathbf{1}$ the imaginary part of the period is an algebraic multiple of π^r .*

Proof. By (??) the period is a combination, with coefficients in $\mathbb{Q}(\zeta_M)$ (the values $w_{\varphi}(e(\text{adj}/c))$ and $\psi(j)$), of $M^{-r}\zeta_H(r, j/M)$, and $\zeta_H(r, j/M) = \frac{M^r}{\phi(M')} \sum_{\chi \bmod M'} \bar{\chi}(j') L(r, \chi)$ with $j/M = j'/M'$ in lowest terms; imprimitive L -values differ from primitive ones by rational Euler factors. The second sentence is Theorem ??(i) below: the imaginary part is a cotangent sum, and $\sum_f \psi(f) f^{-r} \cot(\pi f \theta)$ over a residue class is a combination of $\zeta_H(r, \cdot)$ with coefficients in $\mathbb{Q}(\zeta_M)$ whose real and imaginary parts are, by the parity of ψ and r , multiples of π^r times algebraic numbers (they are values of the Bernoulli-polynomial formula for $L(r, \chi)$ with $\chi(-1) = (-1)^r$). $\square$

Remark 4.3 (representatives). $\Pi_{a/c}^q$ is a function of the rational number a/c , not of its Γ -orbit: $1/N \sim \infty$ under $\Gamma_0(N)$, yet $\Pi_{1/N}^q = L(\Phi, r)$ (for $c = N$ all $\zeta^{df} = 1$ when $N \mid df \dots$; the census gives $\Pi_{1/N}^q = L(\Phi, r)$ for every row, `lattice/cusp_periods/07_ident.gp`, 60 digits) while $\Pi_{\infty}^q = 0$. This is not a contradiction: Θ is not modular, and P_{γ} changes by a period polynomial under $\gamma \mapsto \gamma\delta$, $\delta \in \Gamma$. The monodromy period selects the representative that the radial path in the x -disc actually reaches, and the dictionary of Section ?? records which one that is. The natural statement $\Pi_{\gamma\mathfrak{c}}^q - \Pi_{\mathfrak{c}}^q = (\text{period polynomial of } \Phi \text{ under } \gamma, \text{ evaluated at the cusp})$ is untested.

5. THE ORIENTATION DICHOTOMY

Theorem 5.1 (orientation dichotomy). (i) Inner sources ($\varphi = \mathbf{1}$). *At every cusp a/c ,*

$$\text{Re } \Pi_{a/c}^q = -\frac{1}{2} \sum_d c_d d^{-r} L(r, \psi) = L(\Phi, r), \quad \text{Im } \Pi_{a/c}^q = \frac{1}{2} \sum_d c_d d^{-r} \sum_{f \geq 1, \mathfrak{c} \nmid df} \frac{\psi(f)}{f^r} \cot \frac{\pi a df}{c}.$$

In particular $\Pi_0^q = \Pi_{1/2}^q = L(\Phi, r)$: the orientation factor $\text{Re } \Pi_{\mathfrak{c}}^q / L(\Phi, r)$ is +1 at every cusp. (ii) Outer sources ($\psi = \mathbf{1}$, φ of period Q).

$$\Pi_0^q = \zeta(r) \sum_d c_d d^{-r} \Lambda_{\varphi}(1), \quad \Pi_{1/2}^q = \zeta(r) \sum_d c_d d^{-r} [2^{-r} \Lambda_{\varphi}(1) + (1 - 2^{-r}) \Lambda_{\varphi}((-1)^d)],$$

where $\Lambda_\varphi(1)$ means $w_\varphi(1) = -B_{1,\varphi}$ and $\Lambda_\varphi(-1) = N_\varphi(-1)/(1 - (-1)^Q)$ for Q odd. The ratio $\Pi_{1/2}^q/\Pi_0^q$ is a rational number, in general $\neq 1$.

Proof. (i) $w(z) = z/(1-z) = 1/(z^{-1}-1)$. For $z = e^{i\theta}$, $\theta \neq 0$, $w = \frac{(\cos \theta - 1) + i \sin \theta}{2 - 2 \cos \theta} = -\frac{1}{2} + \frac{i}{2} \cot \frac{\theta}{2}$, and the regularized value at $z = 1$ is $w(1) = -\frac{1}{2}$ too. Insert into (??) with $\theta = 2\pi adf/c$: the real part is $-\frac{1}{2} \sum_d c_d d^{-r} \sum_f \psi(f) f^{-r}$, and $L(\Phi, s) = \sum_d c_d d^{-s} L(s, \psi) \zeta(s-r)$ has the value $\sum_d c_d d^{-r} L(r, \psi) \zeta(0)$ at $s = r$. At $c = 1$ every term has $c \mid df$; at $c = 2$ the terms with df odd have $\cot(\pi df/2) = 0$. (ii) For $\psi = \mathbf{1}$, $\zeta = 1$: every $\zeta^{df} = 1$ and $\Pi_0^q = \sum_d c_d d^{-r} \zeta(r) w_\varphi(1)$. For $\zeta = -1$: $\zeta^{df} = 1$ when df is even and -1 when df is odd; $\sum_{f \text{ even}} f^{-r} = 2^{-r} \zeta(r)$ and $\sum_{f \text{ odd}} f^{-r} = (1 - 2^{-r}) \zeta(r)$. When Q is odd, $(-1)^Q \neq 1$, so $z_0 = -1$ is not a pole and $w_\varphi(-1) = \Lambda_\varphi(-1)$. $\square$

Example 5.2 (the -11 and the -3). *Apéry's $\zeta(2)$ row D:* $\Phi_D = E_3^{1,\varphi}$ with $\varphi = \text{Re } \psi_4 - 2 \text{Im } \psi_4 = (1, -2, 2, -1, 0) \bmod 5$, $r = 2$ [?, Table 1]. Here $N_\varphi(z) = z - 2z^2 + 2z^3 - z^4$, $N_\varphi(1) = 0$, $N'_\varphi(1) = -1$, so $\Lambda_\varphi(1) = \frac{1}{5}$; and $N_\varphi(-1) = -6$, $\Lambda_\varphi(-1) = -6/2 = -3$. Hence $\Pi_0^q = \zeta(2)/5$ and

$$\Pi_{1/2}^q = \zeta(2) \left[\frac{1}{4} \cdot \frac{1}{5} + \frac{3}{4} \cdot (-3) \right] = -\frac{11}{5} \zeta(2), \quad \frac{\Pi_{1/2}^q}{\Pi_0^q} = -11.$$

In polylogarithm language: $D^{-2} \Phi_D = \sum_e \varphi(e) \text{Li}_2(q^e)$, and $\text{Li}_2(1) = \zeta(2)$ while $\text{Li}_2(-1) = -\frac{1}{2} \zeta(2)$, so the weight at $q \rightarrow -1$ is parity-dependent; the regularized character sums are $-\frac{1}{5} \sum_{j \leq 5} j \varphi(j) = \frac{1}{5}$ and $-\frac{1}{10} \sum_{j \leq 10} j(\varphi g)(j) = -\frac{11}{5}$ with $g = (-\frac{1}{2}$ on odd, 1 on even). Nothing here involves $\mathbb{Q}(\sqrt{5})$; that $11 = \varphi^5 - \varphi^{-5} = \text{Tr}(\varphi^5)$ is not claimed to be structural. *The Franel row A:* $\Phi_A = E_3^{1,\chi^{-3}}(\tau) - E_3^{1,\chi^{-3}}(2\tau)$. $\Lambda_{\chi^{-3}}(1) = L(0, \chi_{-3}) = \frac{1}{3}$, $\Lambda_{\chi^{-3}}(-1) = \frac{-2}{2} = -1$; so $\Pi_0^q = \zeta(2)(1 - \frac{1}{4}) \frac{1}{3} = \zeta(2)/4$ and $\Pi_{1/2}^q = \zeta(2) \left[(\frac{1}{12} - \frac{3}{4}) - \frac{1}{4} \cdot \frac{1}{3} \right] = -\frac{3}{4} \zeta(2)$: factor -3 . Both factors are reproduced by the general formula (Verification above) and by monodromy (Section ??).

5.1. Galois equivariance of inner sources. Let $K \subset \mathbb{R}$ be a number field and $\Phi = \sum_j \kappa_j \Phi_j$ with $\kappa_j \in K$ and Φ_j inner sources with rational Fourier coefficients (for instance $\Phi_j = E_{r+1}^{\psi,1} + E_{r+1}^{\bar{\psi},1}$ and $i(E_{r+1}^{\psi,1} - E_{r+1}^{\bar{\psi},1})$). For an embedding $\sigma : K \rightarrow \mathbb{R}$ put $\sigma\Phi := \sum_j \sigma(\kappa_j) \Phi_j$.

Corollary 5.3. $\text{Re } \Pi_{\mathfrak{c}}^q(\sigma\Phi) = \sum_j \sigma(\kappa_j) L(\Phi_j, r)$ for every cusp $\mathfrak{c}$. In words: the real cusp period of $\sigma\Phi$ at any cusp is obtained from the real cusp period of Φ at any other cusp by applying σ to the K -coefficients in its expression through the fixed numbers $L(\Phi_j, r)$.

Proof. $\Pi_{\mathfrak{c}}^q$ is linear in the source, and by Theorem ??(i) $\text{Re } \Pi_{\mathfrak{c}}^q(\Phi_j) = L(\Phi_j, r)$ is independent of $\mathfrak{c}$. $\square$

For outer sources this fails: the real periods carry a cusp-dependent rational factor, -11 on the $\zeta(2)$ -line of $\Gamma_1(5)$, and a K -rational outer source would conjugate to a source whose far period is σ of the near period times that factor.

5.2. The example $\Gamma_1(5)$ in full. The cusps of $\Gamma_1(5)$ are ∞ , $2/5$ (width 1) and 0 , $1/2$ (width 5), with $x(\infty) = 0$, $x(0) = t_1 = \varphi^{-5}$, $x(1/2) = t_2 = -\varphi^5$, $x(2/5) = \infty$ (lattice/gamma15/task2/13_haupt.gp, 38 digits); $t_1 t_2 = -1$. The weight-three Eisenstein space is four-dimensional, spanned by the outer $E_3^{1,\psi_4}, E_3^{1,\bar{\psi}_4}$ and the inner $E_3^{\psi_4,1}, E_3^{\bar{\psi}_4,1}$, ψ_4 the quartic character with $\psi_4(2) = i$. The source with a period at t_1 must have $a_0 = 0$ at ∞ and at 0 (Proposition ??, Theorem ??); in the inner pair this forces, by Corollary ??(i) at $c = 1$,

$$(8) \quad \alpha L(3, \psi_4) + \bar{\alpha} L(3, \bar{\psi}_4) = 0 \quad \text{for } \Phi = \alpha E_3^{\psi_4,1} + \bar{\alpha} E_3^{\bar{\psi}_4,1},$$

which by Proposition ?? below has the solution $\alpha = 1 + i\varphi^5$: $\Phi_{\text{new}} = (1 + i\varphi^5)E_3^{\psi_4,1} + (1 - i\varphi^5)E_3^{\bar{\psi}_4,1} = 2q + (2 - 10\varphi)q^2 + (24 + 10\varphi)q^3 + \dots \in \mathbb{Z}[\varphi][[q]]$. Its periods, by Theorem ??(i):

$$\Pi_0^q(\Phi_{\text{new}}) = \Pi_{1/2}^q(\Phi_{\text{new}}) = -\frac{1}{2}[(1+i\varphi^5)L(2, \psi_4) + (1-i\varphi^5)L(2, \bar{\psi}_4)] = \varphi^5 \operatorname{Im} L(2, \psi_4) - \operatorname{Re} L(2, \psi_4) = \xi.$$

The conjugate source $\sigma\Phi_{\text{new}} = (1-i\varphi^{-5})E_3^{\psi_4,1} + (1+i\varphi^{-5})E_3^{\bar{\psi}_4,1}$ has $\Pi_0^q = \Pi_{1/2}^q = -\operatorname{Re} L(2, \psi_4) - \varphi^{-5} \operatorname{Im} L(2, \psi_4) = \sigma(\xi)$. Which of these are monodromy periods is decided by the polar coefficients: $\rho_0(\Phi_{\text{new}}) = 0$ by (??), while $\rho_{1/2}(\Phi_{\text{new}}) = \frac{1}{8}[i(1+i\varphi^5)L(3, \psi_4) - i(1-i\varphi^5)L(3, \bar{\psi}_4)] = -2.7620797186\dots \neq 0$; for $\sigma\Phi_{\text{new}}$ the roles are exchanged (since $\psi_4(2) = i$ turns $\alpha \mapsto i\alpha$, and $\sigma(\varphi^5) = -\varphi^{-5}$). So

(9)

$$\Pi_{t_1}(\Phi_{\text{new}}) = \xi, \quad \Pi_{t_2}(\sigma\Phi_{\text{new}}) = \sigma(\xi), \quad \text{and} \quad \Pi_{t_1}(\Phi_D) = \frac{1}{5}\zeta(2), \quad \Pi_{t_2}(\Phi_D) = -\frac{11}{5}\zeta(2),$$

while B_{new} has a $\log^2$ at t_2 and σB_{new} a $\log^2$ at t_1 (Lemma ??(ii): $\mathcal{L}B_{\text{new}} = 2/(1-x/t_2)$, $\mathcal{L}\sigma B_{\text{new}} = 2/(1-x/t_1)$, exact identities of rational functions verified on 198 coefficients, `lattice/gamma15/task2/`). All four values in (??) were confirmed by numerical monodromy of the row operator to 174 digits at t_2 and 212 digits at t_1 (`lattice/gamma15/task2/17_farperiods.gp`, `18_nearperiods.gp`) and to 77 digits from the cusp formula at all four cusps (`lattice/cusp_periods/08_galois.gp`).

Proposition 5.4 (the $L(3, \psi_4)$ identity). $\operatorname{Re} L(3, \psi_4) = \varphi^5 \operatorname{Im} L(3, \psi_4)$; equivalently (??) holds for $\alpha = 1 + i\varphi^5$.

Proof. For a primitive odd character χ modulo N , $L(3, \chi) = -i(2\pi)^3 \tau(\chi) B_{3, \bar{\chi}} / (12N^3)$ with $B_{3, \bar{\chi}} = N^2 \sum_{b=1}^N \bar{\chi}(b) B_3(b/N)$, $B_3(x) = x^3 - \frac{3}{2}x^2 + \frac{1}{2}x$; this follows from $\sum_{m \geq 1} \sin(2\pi mx)/m^3 = (2\pi)^3 B_3(x)/12$ for $0 \leq x \leq 1$ and finite Fourier inversion $\chi(m) = \frac{\tau(\chi)}{N} \sum_b \bar{\chi}(b) e(-mb/N)$. For $N = 5$: $B_3(\frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}) = \frac{1}{125}(6, 3, -3, -6)$, and $\bar{\psi}_4 = (1, -i, i, -1)$ on $1, 2, 3, 4$ gives $B_{3, \bar{\psi}_4} = 25 \cdot \frac{12-6i}{125} = \frac{12-6i}{5}$; and $\tau(\psi_4) = e(\frac{1}{5}) + ie(\frac{2}{5}) - ie(\frac{3}{5}) - e(\frac{4}{5}) = 2is_1 - 2s_2$ with $s_1 = \sin \frac{2\pi}{5}$, $s_2 = \sin \frac{\pi}{5}$. Multiplying out,

$$\operatorname{Re} L(3, \psi_4) = \frac{16\pi^3}{7500}(12s_1 + 6s_2), \quad \operatorname{Im} L(3, \psi_4) = \frac{16\pi^3}{7500}(12s_2 - 6s_1),$$

so the ratio is $(2s_1 + s_2)/(2s_2 - s_1)$. Since $s_1/s_2 = 2 \cos \frac{\pi}{5} = \varphi$, this is $(2\varphi + 1)/(2 - \varphi) = \varphi^3/\varphi^{-2} = \varphi^5$, using $\varphi^3 = 2\varphi + 1$ and $\varphi^{-2} = 2 - \varphi$. For the equivalence, $\alpha L(3, \psi_4) + \bar{\alpha} \overline{L(3, \psi_4)} = 2 \operatorname{Re}(\alpha L(3, \psi_4)) = 2(\operatorname{Re} L - \varphi^5 \operatorname{Im} L)$. $\square$

Note that the naive ratio $B_{3, \operatorname{Re} \psi_4} / B_{3, \operatorname{Im} \psi_4} = 2$ is *not* φ^5 : the Gauss sum does not commute with real and imaginary parts. Numerically the identity holds to 76 digits (`check/02_L3identity.gp`), 96 digits (`lattice/cusp_periods/03_L3exact.gp`) and 212 digits (`lattice/gamma15/L3_identity.gp`). It is the $s = 3$ shadow of the fold-regularity ratio $-\varphi^5$ between the inner directions at $s = 2$, and it is what makes the $\sqrt{5}$ in Φ_{new} unavoidable.

Remark 5.5 (weight four on $\Gamma_1(5)$). At weight four the four Eisenstein directions $E_4(\tau), E_4(5\tau), E_4^{\chi_5,1}, E_4^{1, \chi_5}$ have real cusp periods $-\zeta(3)/2, -\zeta(3)/250, -L(3, \chi_5)/2, 0$ and constant terms $\frac{1}{240}, \frac{1}{240}, 0, 1$ at ∞ : everything, including the fold-regular subspace, is $\mathbb{Q}$ -rational (`lattice/cusp_periods/08_galois.gp`, 60 digits). Beukers' $\sqrt{5}$ in $8\zeta(3) - 5\sqrt{5}L(3, \chi_5)$ [?] is thus geometry, not Eisenstein bookkeeping; weight three is the only weight on $\Gamma_1(5)$ where φ^5 is forced.

6. THE CENSUS, AND ITS CONFIRMATION BY MONODROMY

6.1. Near periods and all cusps. Table ?? lists, for the twelve Eisenstein rows of [?, Table 1], the near period $L(\Phi, r) = \Pi_0^q$ computed from (??) alone, and at every cusp of $\Gamma_0(N_\Phi)$ (the level of the source) whether $\rho_c = 0$ (*fr*), the orientation factor $\operatorname{Re} \Pi_c^q / L(\Phi, r)$, and $\operatorname{Im} \Pi_c^q$. The near

periods reproduce the Apéry-limit column of [?] in every case; this is an independent derivation of that column from the cusp side. Every cusp value is $(\text{rational}) \cdot L(\Phi, r) + (\text{rational}) \cdot \pi^r / \sqrt{D} \cdot i$, and a factor $\neq 1$ occurs only for **A** (-3) , **D** (-11) and $\zeta(-\frac{1}{2})$, at non-regular cusps) — exactly the three sources with $\varphi \neq 1$, as Theorem ?? predicts. Rows **B** (level 36, eleven cusps), δ and η have factor 1 at every cusp with imaginary parts in $\mathbb{Q}\pi^2/\sqrt{3}$, $\mathbb{Q}\pi^3$, $\mathbb{Q}\pi^3/\sqrt{5}$ respectively.

TABLE 1. Cusp census (`lattice/cusp_periods/02_census.gp`, `07_ident.gp`, 60 digits). Columns: near period; then for each cusp $\mathfrak{c}$ of $\Gamma_0(N_\Phi)$: fold-regular?, factor $\text{Re } \Pi_{\mathfrak{c}}^q / L(\Phi, r)$, $\text{Im } \Pi_{\mathfrak{c}}^q$.

row	$L(\Phi, r)$	orientation	cusps: $\mathfrak{c}$ (fr, factor, Im)
A	$\zeta(2)/4$	outer χ_{-3}	$0(\checkmark, 1, 0); \frac{1}{2}(\checkmark, -\mathbf{3}, 0); \frac{1}{3}(\times, -\frac{1}{3}, 0); \frac{1}{6}(\checkmark, 1, 0)$
C	$L(2, \chi_{-3})/2$	inner χ_{-3}	$0(\checkmark, 1, 0); \frac{1}{2}(\times, 1, 0); \frac{1}{3}(\checkmark, 1, \frac{2\pi^2}{9\sqrt{3}}); \frac{1}{6}(\checkmark, 1, 0)$
D	$\zeta(2)/5$	outer	$0(\checkmark, 1, 0); \frac{1}{2}(\checkmark, -\mathbf{11}, 0); \frac{2}{5}(\times, -\frac{7}{5}, 0)$ [on $\Gamma_1(5)$]
E	$G/2$	inner χ_{-4}	$0(\checkmark, 1, 0); \frac{1}{2}(\times, 1, 0); \frac{1}{4}(\checkmark, 1, \frac{\pi^2}{16}); \frac{1}{8}(\checkmark, 1, 0)$
F	$\frac{5}{8}L(2, \chi_{-3})$	inner χ_{-3}	$0(\checkmark, 1, 0); \frac{1}{2}(\times, 1, 0); \frac{1}{3}(\checkmark, 1, \frac{\pi^2}{6\sqrt{3}}); \frac{1}{4}(\checkmark, 1, \frac{\pi^2}{12\sqrt{3}}); \frac{1}{6}(\checkmark, 1, \frac{\pi^2}{18\sqrt{3}})$
α	$7\zeta(3)/24$	trivial	$0(\checkmark, 1, 0); \frac{1}{2}(\times, 1, 0); \frac{1}{3}(\checkmark, 1, \frac{\pi^3}{36}); \frac{1}{4}(\checkmark, 1, \frac{\pi^3}{48}); \frac{1}{6}(\times, 1, \frac{\pi^3}{108})$
γ	$\zeta(3)/6$	trivial	$0(\checkmark, 1, 0); \frac{1}{2}(\times, 1, 0); \frac{1}{3}(\times, 1, \frac{\pi^3}{27}); \frac{1}{6}(\checkmark, 1, 0)$
ε	$7\zeta(3)/32$	trivial	$0(\checkmark, 1, 0); \frac{1}{2}(\times, 1, 0); \frac{1}{4}(\times, 1, \frac{\pi^3}{64}); \frac{1}{8}(\checkmark, 1, 0)$
ζ	$L(3, \chi_{-3})/3$	χ_{-3} both	$0(\checkmark, 1, 0); \frac{1}{3}, \frac{2}{3}(\times, -\frac{1}{2}, 0); \frac{1}{9}(\checkmark, 1, 0)$
B, δ, η	$\frac{1}{2}L(2, \chi_{-3}), \frac{13}{54}\zeta(3), \frac{1}{2}L(3, \chi_5)$	inner, trivial, inner	factor 1 at every cusp

On $\Gamma_1(5)$ the full table is: Φ_D : $0, \zeta(2)/5, -\frac{11}{5}\zeta(2), -\frac{7}{25}\zeta(2)$ at $\infty, 0, \frac{1}{2}, \frac{2}{5}$ (fold-regular at the first three); Φ_{new} : $0, \xi, \xi, \xi + 7.0335782\dots i$ (fold-regular at $\infty, 0, \frac{2}{5}$); $\sigma\Phi_{\text{new}}$: $0, \sigma\xi, \sigma\xi, \sigma\xi + 0.3171086\dots i$ (fold-regular at $\infty, \frac{1}{2}, \frac{2}{5}$); the imaginary parts at the width-one cusp $2/5$, which maps to $x = \infty$, were not identified (`12_imid.gp`).

6.2. The monodromy dictionary. The claim that $\Pi_{\mathfrak{c}}^q$ is the period was tested directly: the row operator $\mathcal{L}$ (second order, $P(x) = 1 - ax + cx^2$: $xPy'' + (P + xP')y' + (cx - b)y = R$; third order analogously) was continued numerically from 520–620 exact rational Taylor coefficients at $x = 0$ around each finite singular point (adaptive Taylor stepping, 260 terms per step, 100–120-digit working precision, `lattice/cusp_periods/mono.gp`), and the ratio $\ell(B)/\ell(A)$ of the monodromy differences $(M - 1)B, (M - 1)A$ was computed independently from y and y' . Table ?? gives the results and the cusp that the q -side formula must be evaluated at to match.

TABLE 2. Monodromy periods at both finite singular points against the cusp formula (`05_mono.gp`, `06_mono3.gp`).

row	singular point	monodromy period	cusp	difference
A	$\frac{1}{8} / -1$	$\zeta(2)/4 / -\frac{3}{4}\zeta(2)$	$0 / \frac{1}{2}$	$3.5 \cdot 10^{-135} / 6.0 \cdot 10^{-134}$
C	$\frac{1}{9} / 1$	$\frac{1}{2}L(2, \chi_{-3}) / \frac{1}{2}L(2, \chi_{-3}) + \frac{2\pi^2}{9\sqrt{3}}i$	$0 / \frac{1}{3}$	$1.6 \cdot 10^{-135} / 1.1 \cdot 10^{-119}$
E	$\frac{1}{8} / \frac{1}{4}$	$G/2 / G/2 + \frac{\pi^2}{16}i$	$0 / \frac{1}{4}$	$3.3 \cdot 10^{-135} / 6.7 \cdot 10^{-120}$
F	$\frac{1}{9} / \frac{1}{8}$	$\frac{5}{8}L(2, \chi_{-3}) / \frac{5}{8}L(2, \chi_{-3}) + \frac{\pi^2}{18\sqrt{3}}i$	$0 / \frac{1}{6}$	$4.1 \cdot 10^{-120} / 4.0 \cdot 10^{-120}$
D	$\varphi^{-5} / -\varphi^5$	$\zeta(2)/5 / -\frac{11}{5}\zeta(2)$	$0 / \frac{1}{2}$	$1.6 \cdot 10^{-135} / 2.9 \cdot 10^{-121}$
α	$\frac{1}{16} / \frac{1}{4}$	$7\zeta(3)/24 / 7\zeta(3)/24 + \frac{\pi^3}{48}i$	$0 / \frac{1}{4}$	$< 10^{-100} / < 10^{-100}$

The imaginary parts are reproduced too: they are the cotangent sums of Theorem ??(i), not artefacts. Note that for $\mathbf{F}$ the far singularity $x = \frac{1}{8}$ is the cusp $\frac{1}{6}$, not $\frac{1}{2}$: this is the representative caveat of Remark ?? in action. Rows $\mathbf{B}, \delta, \eta$ have complex conjugate finite singularities and were not continued.

6.3. Cooper's meromorphic rows. The sources of s_7, s_{10}, s_{18} are meromorphic (double poles at CM points, [?, §3.5]), so Theorem ?? does not apply; continuation alone gives (09_cooper_gp, 100 digits) near periods $\zeta(2)/7, \zeta(2)/5, \frac{1}{2}L(2, \chi_{-3})$ at $\frac{1}{27}, \frac{1}{16}, \frac{1}{16}$ and far-fold periods $-\frac{5}{7}\zeta(2)$ at $-1, -\frac{2}{5}\zeta(2)$ at $-\frac{1}{4}$, and $\frac{1}{2}L(2, \chi_{-3}) + \frac{\pi^2}{18\sqrt{3}}i$ at $\frac{1}{12}$: factors $-5, -2, +1$. The two rows whose companion character is $\psi = \mathbf{1}$ [?, Thm. 3.3] behave like outer rows and the one with $\psi = \chi_{-3}$ like an inner row; and s_{18} 's far imaginary part coincides with Zagier $\mathbf{F}$'s to all 100 digits.

7. CUSP-FORM SOURCES ON FRICKE HOSTS

The companion formula (??) gives $d_n^r B_n \in \mathbb{Z}$ for *any* integral source; Eisenstein sources are needed only for regularity. On the Fricke hosts of the third-order rows the regularity condition at the near singularity, which is a fold, has a clean answer. Let $u = q + O(q^2)$ be an eta quotient with $u|W_N = 1/(Cu)$, $x = u/(1 + Bu + Cu^2)$, $F = D \log u$ (weight 2, $F|_2 W_N = -F$), as in [?, (2.5)]; x is W_N -invariant and has a simple critical point at $\tau_c = i/\sqrt{N}$ with $x(\tau_c) = 1/\lambda_1$ the nearest singularity, and $r = 3$.

Theorem 7.1 (Fricke criterion). *Let $\Phi \in M_4(\Gamma_0(N))$ with $a_0(\Phi, \infty) = 0$ and $\Theta = D^{-3}\Phi$. Then $B = F\Theta$ has a period at $x(\tau_c)$ if and only if $\Phi|_4 W_N = -\Phi$, and then the period is $\xi = L(\Phi, 3) = 4\pi^3 \Lambda(3)$, $\Lambda(j) := \int_0^\infty y^{j-1} \Phi(iy) dy$. Moreover, for any such Φ and any host of level N ,*

$$\xi = \Theta(q_c) + \frac{2\pi}{\sqrt{N}}(D\Theta)(q_c), \quad q_c = e^{-2\pi/\sqrt{N}}.$$

Proof. Since $x \circ W_N = x$, W_N fixes τ_c and acts near it as the non-trivial deck transformation of the double cover $\tau \mapsto x$; a function analytic near τ_c is analytic in x near $x(\tau_c)$ iff it is W_N -invariant. Now $F(W_N \tau) = -N\tau^2 F(\tau)$ and $\Theta(W_N \tau) = N^{-1}\tau^{-2}(\Theta|_{-2} W_N)(\tau)$, so

$$[F(\Theta - \xi)] \circ W_N = -F[\Theta|_{-2} W_N - N\tau^2 \xi],$$

and $B - \xi F = F(\Theta - \xi)$ is W_N -invariant iff

$$(10) \quad \Theta|_{-2} W_N + \Theta = \xi(N\tau^2 + 1).$$

With $\Theta = \frac{(2\pi i)^3}{2} \int_{i\infty}^\tau (\tau - v)^2 \Phi(v) dv$ and $\Phi(-1/(Nv)) = \varepsilon N^2 v^4 \Phi(v)$ for $\Phi|_4 W_N = \varepsilon \Phi$, the substitution $u = -1/(Nv)$ gives $\Theta|_{-2} W_N = \varepsilon(\Theta + P_0)$ with $P_0(\tau) = \frac{(2\pi i)^3}{2} \int_0^{i\infty} (\tau - v)^2 \Phi(v) dv = 4\pi^3 [\Lambda(1)\tau^2 - 2i\Lambda(2)\tau - \Lambda(3)]$. If $\Phi = \Phi_+ + \Phi_-$ in W_N -eigencomponents, (??) reads $2\Theta_+ + P_0^+ - P_0^- = \xi(N\tau^2 + 1)$, forcing Θ_+ to be a polynomial, i.e. $\Phi_+ = 0$. For $\varepsilon = -1$, (??) reads $-P_0 = \xi(N\tau^2 + 1)$: $\Lambda(2) = 0$, $4\pi^3 \Lambda(3) = \xi$, $-4\pi^3 \Lambda(1) = N\xi$. The functional equation $\Lambda(s) = \varepsilon N^{2-s} \Lambda(4-s)$ (from $\Phi(i/(Ny)) = \varepsilon N^2 y^4 \Phi(iy)$) gives $\Lambda(2) = 0$ and $\Lambda(1) = -N\Lambda(3)$ automatically, so (??) holds with $\xi = 4\pi^3 \Lambda(3) = (2\pi)^3 \Gamma(3)^{-1} \cdot \Gamma(3)(2\pi)^{-3} L(\Phi, 3) \cdot 4\pi^3 / (4\pi^3) = L(\Phi, 3)$ (indeed $\Lambda(3) = \Gamma(3)(2\pi)^{-3} L(\Phi, 3) = L(\Phi, 3)/(4\pi^3)$). Finally differentiate (??) at τ_c , where $N\tau_c^2 = -1$: $\frac{d}{d\tau}[N\tau^2 \Theta(-1/(N\tau))] = 2N\tau \Theta(W_N \tau) + \Theta'(W_N \tau)$, so $2N\tau_c \Theta(\tau_c) + 2\Theta'(\tau_c) = 2\xi N\tau_c$, i.e. $\xi = \Theta(\tau_c) + \Theta'(\tau_c)/(N\tau_c) = \Theta(\tau_c) + \frac{2\pi}{\sqrt{N}} D\Theta(\tau_c)$. $\square$

For $\varepsilon = +1$ the limit B_n/A_n still exists but converges like $\xi + c/n$ and is not $L(\Phi, 3)$ (measured ratio 0.99000 against $((n-1)/n)^2$ at $n = 200$ on all twelve hosts, lattice/cuspform_sources/10_plus_gp); no Eisenstein admixture repairs it, since $M_4 = \text{Eis} \oplus S_4$ is W_N -stable. The census (01_forms_gp, exact): $\dim S_4(\Gamma_0(N))^{W_N=-1} = 0$ for $N = 5, \dots, 9$ — the newforms 5.4.a.a, $\dots$, 9.4.a.a all have

Fricke sign +1 — so no cusp form is regular on either Apéry-perfect host ($N = 6, 7$); the anti-invariant directions that exist are oldform combinations $f - d^2 f(d\tau)$, and there are five (host, source) pairs, Table ???. Where the constant K_- of $A_n \xi - B_n \sim K_- \lambda_2^n n^{-3/2}$ was identified (`11_kminus.gp`, `14_kclosed.gp`, 50 digits), it is the *central* value $L(f, 2)$, non-zero because the newform’s own sign is +1; we record this as an observation. The companions satisfy no three-term recurrence (minimal order 3 or 4), so the Casoratian formula of [?, Thm. 4.5] does not apply. The Diophantine content is nil: the best host misses the entry condition by 4.4 nats.

TABLE 3. Fold-regular cusp-form sources on Fricke hosts
(`lattice/cuspform_sources/`, limits to 144 digits on the first two rows).

host (N, B, C)	source	ξ	rate λ_2/λ_1	K_-
Cooper s_{10} (10, 6, 25)	$f_5 - 4f_5(2\tau)$	$L(5.4.a.a, 3)/2$	1/4	not identified
Domb (12, 10, 9)	$f_6 - 4f_6(2\tau)$	$L(6.4.a.a, 3)/2 = 0.364556728783\dots$	1/4	$L(6.4.a.a, 2)/\sqrt{\pi}$
level 12, (12, 34, 1)	$f_6 - 4f_6(2\tau)$	$L(6.4.a.a, 3)/2$	8/9	not identified
Cooper s_{18} (18, 14, 1)	$f_6 - 9f_6(3\tau)$	$2L(6.4.a.a, 3)/3$	3/4	$\sqrt{3}L(6.4.a.a, 2)/\sqrt{\pi}$
Cooper s_{18} (18, 14, 1)	$f_9 - 4f_9(2\tau)$	$L(9.4.a.a, 3)/2$	3/4	not identified

8. PERIODS AT FOLDS

[Placeholder: a section on fold periods, with theorems from the period-matrix computation, is being prepared separately and will replace this paragraph.] For the rows γ (Apéry’s $\zeta(3)$), ε and ζ the far singularity is a second W_N -fixed point, not a cusp, and its period matches no cusp of the source level. By continuation (`lattice/cusp_periods/06_mono3.gp`, `09_cooper.gp`, 100 digits) the far periods are

$$\gamma : \frac{1}{6}\zeta(3) + \frac{2\pi^3}{15}i \text{ at } 17+12\sqrt{2}, \quad \varepsilon : \frac{7}{32}\zeta(3) + \frac{\pi^3}{24}i \text{ at } \frac{3+2\sqrt{2}}{4}, \quad \zeta : -\frac{7}{2} \cdot \frac{1}{3}L(3, \chi_{-3}) \text{ at } -0.71823351279\dots,$$

against cusp values $\frac{1}{6}\zeta(3) + \frac{\pi^3}{27}i$ (cusp $\frac{1}{3}$), $\frac{7}{32}\zeta(3) + \frac{\pi^3}{64}i$ (cusp $\frac{1}{4}$) and factors $1, -\frac{1}{2}$: the ratios $\frac{18}{5}, \frac{8}{3}$ and the factor $-\frac{7}{2}$ have no q -side derivation here. The near-fold periods equal $L(\Phi, r) = \Pi_0^q$ in all three cases, which is the content of Theorem ?? and of [?, Thm. B], not of Theorem ??. The fold analogue of Theorem ?? is a W_N -twisted Mellin split of the Eichler integral at τ_c ; the constant $\xi = \Theta(q_c) + \frac{2\pi}{\sqrt{N}}D\Theta(q_c)$ of Theorem ?? is its near-fold instance.

9. CONSEQUENCES FOR LINEAR-INDEPENDENCE PROOFS

Let K be a real quadratic field and suppose one wants to prove that $1, \zeta(2), \xi$ are linearly independent over $K = \mathbb{Q}(\sqrt{5})$ by the arithmetic-holonomy method of [?] run over K , on Apéry’s $\zeta(2)$ host with the two-dimensional fold-regular source space $\langle \Phi_D, \Phi_{\text{new}} \rangle$. The method builds, from a hypothesized relation

$$(i) \quad a + b\zeta(2)/5 + c\xi = 0, \quad a, b, c \in K,$$

a conditional function $H = aA + bB_D + cB_{\text{new}}$ that is regular at the fold at the first real place, and needs its conjugate σH to be regular at the fold at the second place. The Picard–Fuchs operator has rational coefficients, so its singular set $\{0, \varphi^{-5}, -\varphi^5, \infty\}$ is the same at both places, but the puncture of the pure module, $s = 1/\lambda_2 = -\varphi^5 \in K$, moves: the point to be removed at the second place is $t_2 = -\varphi^5$, the image of the cusp $1/2$. There the periods are those of (??), and regularity of σH is

$$(ii) \quad \sigma(a) - 11\sigma(b)\zeta(2)/5 + \sigma(c)\sigma(\xi) = 0.$$

By Theorem ??, (ii) is σ applied to the K -coefficients of (i) *except* in the $\zeta(2)$ slot, where the outer orientation of Φ_D inserts the factor -11 ; on the inner ξ -line the equivariance is exact. Consequently (i) does not imply (ii) whenever $c \neq 0$: writing (i) as $\Lambda \cdot (1, \zeta(2)/5, \operatorname{Re} L, \operatorname{Im} L) = 0$ with $\Lambda = (a, b, -c, c\varphi^5)$, (ii) is $\Lambda' = \sigma(\Lambda) - 12\sigma(b)e_2$, and $\Lambda' \in K^\times \Lambda$ forces $\sigma(\varphi^5) = \varphi^5$ or $c = 0$. So a single K -relation buys a conditional function at one place only, and the construction refutes the doubled hypothesis (i) $\wedge$ (ii) — two independent $\mathbb{Q}$ -relations among $\{1, \zeta(2), \operatorname{Re} L, \operatorname{Im} L\} \cdot \{1, \sqrt{5}\}$ — rather than (i). This *hypothesis tax* is therefore an orientation theorem: it is absent exactly when the source is inner at both cusps involved, and a two-place construction over a real quadratic field has a Galois-equivariant hypothesis iff its fold-regular space is spanned by inner directions at the near and at the far cusp. (For the record, the transported bound on $\Gamma_1(5)$ falls short by 1.44 nats on (i) $\wedge$ (ii) and by 4.40 nats on (i) alone, for geometric reasons at the second place that are independent of the orientation; the accounting is in `lattice/gamma15/REPORT.md`.)

Two further consequences. *Fold-regularity is computable exactly*: it is the joint kernel of the constant-term functionals at ∞ and at the near cusp (Theorem ??, Proposition ??), all of which are algebraic multiples of π^{r+1} (critical values, Corollary ??); the far period of a fold-regular source is a twisted $L(r)$ -value with an explicit root-of-unity weight (Corollary ??), which is all the Diophantine information the far cusp carries. *One recurrence, many periods*: each fold-regular cusp $\mathfrak{c}$ gives an exact linear form $A_n \Pi_{\mathfrak{c}} - B_n^{(c)}$ in $\Pi_{\mathfrak{c}}$ with the same d_n^r denominators, but only the one at the nearest singularity converges.

10. OPEN QUESTIONS

- (1) The representative statement of Remark ??: $\Pi_{\gamma_{\mathfrak{c}}}^q - \Pi_{\mathfrak{c}}^q$ should be the period polynomial of Φ under γ evaluated at the cusp.
- (2) The fold analogue of Theorem ??, which would derive $\frac{2\pi^3}{15}$, $\frac{\pi^3}{24}$ and $-\frac{7}{2}$ of Section ??.
- (3) The imaginary parts of the periods of Φ_{new} at the width-one cusps of $\Gamma_1(5)$ (7.0335782469643621718..., 0.3171086774431510891...), presumably in $\mathbb{Q}(\zeta_5, \varphi)\pi^2$; `lindep` against $\pi^2\{1, 5^{1/4}, \sqrt{5}, 5^{3/4}\}$ finds nothing at 40 digits.
- (4) The cusp-to- x dictionary for the rows $\mathbf{B}, \delta, \eta$ with complex conjugate singularities, not verified by continuation.
- (5) A host whose *outer* direction has uniform cusp weights (Lambert rather than polylogarithm orientation) would remove the hypothesis tax of Section ??; the host sweep of the project found no Apéry-perfect host at levels ≤ 120 other than 5 and 6.

11. COMPUTATIONS

All computations are in PARI/GP 2.15. The cusp formula, the census, the polar identity, the monodromy dictionary and the Galois test are in `lattice/cusp_periods/` (18 scripts, `REPORT.md`); the $\Gamma_1(5)$ far-cusp periods in `lattice/gamma15/task2/`; the new period ξ and its source in `lattice/hostscan/` (§10 of its report, 128 digits); the cusp-form census in `lattice/cuspform_sources/`. The independent re-derivations for this paper are in `paper/cusp_periods/check/`: `01_cuspformula.gp` (the formula (??) implemented from scratch: $\zeta(2)/5$, $-\frac{11}{5}\zeta(2)$, $\zeta(2)/4$, $-\frac{3}{4}\zeta(2)$, ξ , $\sigma(\xi)$, and the imaginary parts of $\mathbf{C}$ at $\frac{1}{3}$ and $\mathbf{E}$ at $\frac{1}{4}$, all to 76 digits), `02_L3identity.gp` (Proposition ??), `03_showcase.gp` (the showcase page).

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